DHBW Ravensburg

Intelligence Engineering

Solving Real World Problems At Scale

Prof. Dr.-Ing. Mark Schutera | October 26, 2026 · *drifting stream*

LECTURE II

Introduction to Machine Learning Systems Design

This lecture in one slide

II Introduction to Machine Learning Systems Design

Business and ML Objectives
Requirements for ML Systems
Iterative Process
Framing ML Problems
Mind Versus Data

why this lecture matters, and what you can do after it

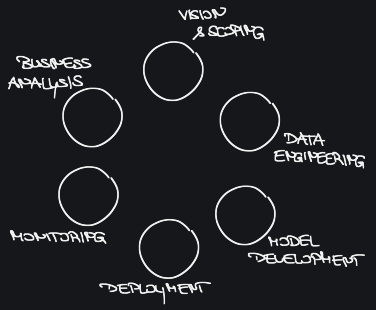
1

Business and ML Objectives

System Design

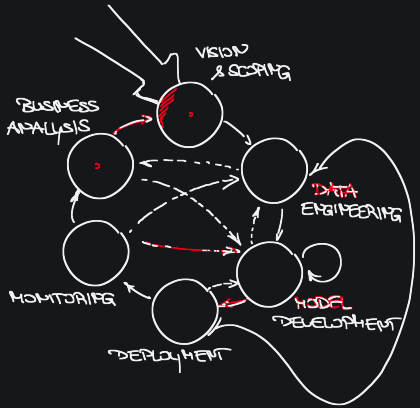
ML systems design takes a **system approach** to MLOps: the business requirements, the data stack, infrastructure, deployment, monitoring, and their stakeholders work together to satisfy the specified objectives and requirements.

If this system is built for a business, it must be driven by business objectives, which will need to be translated into ML objectives to guide the development of ML models.



[1]

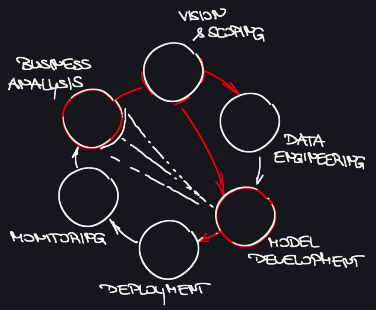
It is not crazy, it is just complex



Tie ML to business

For an ML project to succeed within a business organization, it's crucial to tie the performance of an ML system to the overall business performance. At best the ML metric becomes a **proxy** for a business metric.

- **Ad click-through rate** every increase results in actual ad revenue.
- **Fraud detection** every fraudulent transaction stopped results in actual money saved.



[1]

2

Requirements for ML Systems

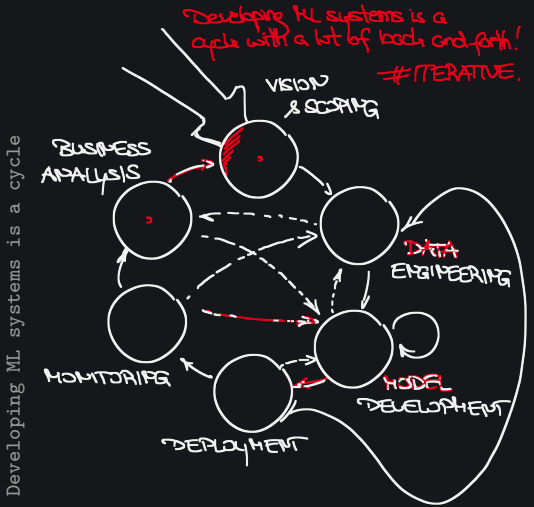
The four requirements

- **RELIABILITY**
  Correct function at desired level of performance in the face of adversity.
- **SCALABILITY**
  With respect to complexity, throughput, and other types of growth.
- **MAINTAINABILITY**
  Structure workload, infrastructure and tool landscape so that you can keep up.
- **ADAPTABILITY**
  Capacity to discover potentials for performance improvement and allowing updates without service interruptions.

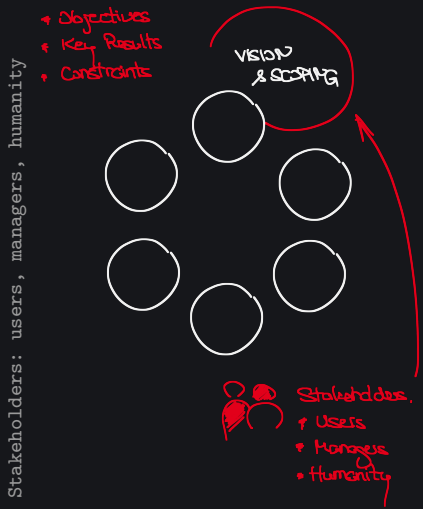
3

Iterative Process

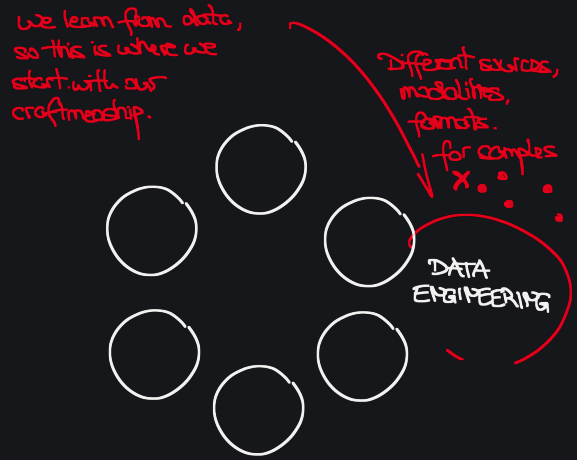
Developing ML systems is a cycle



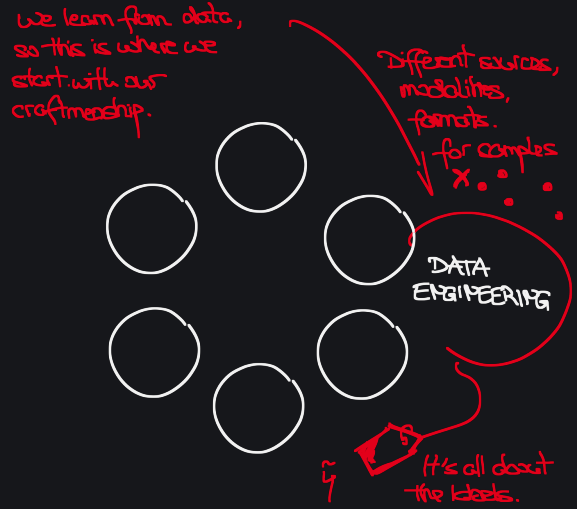
Vision and scoping


Stakeholders: users, managers, humanity
• Objectives
• Key Results
• Constraints
VISION & SCOPING
Stakeholders.
• Users
• Managers
• Humanity

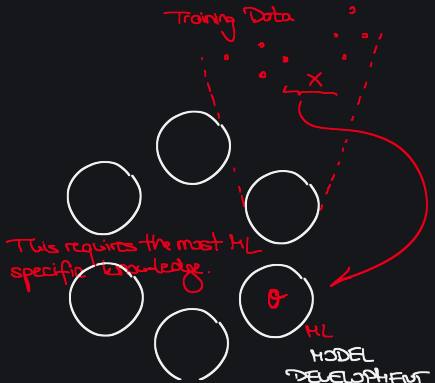
Data engineering: the samples



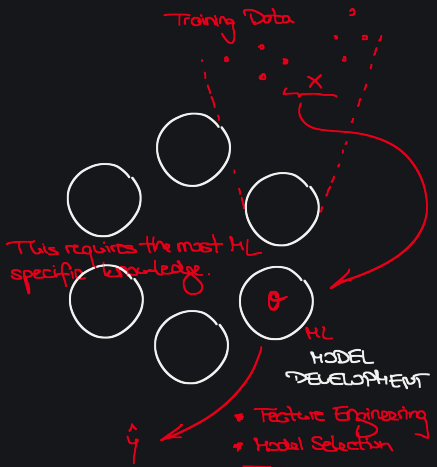
It is all about the labels



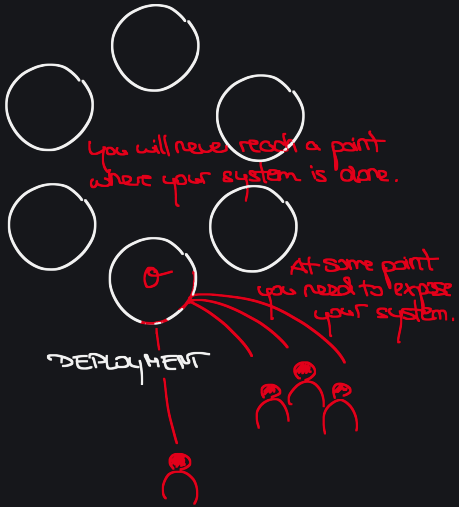
Model development



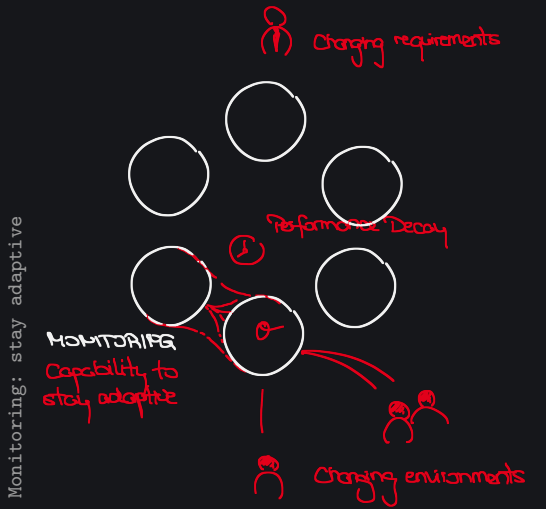
Feature engineering to evaluation



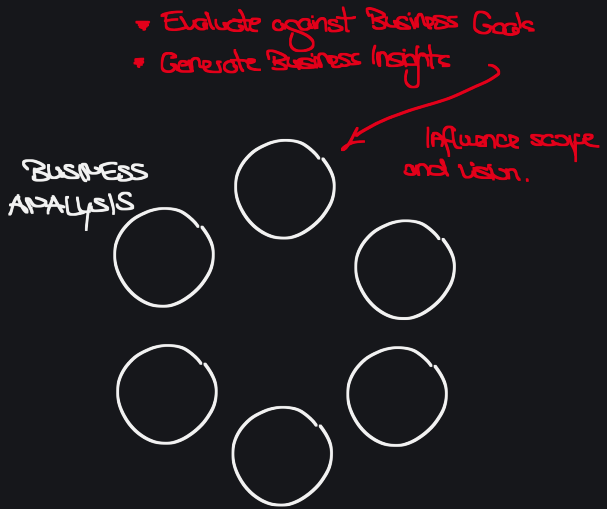
Deployment: expose your system



Monitoring: stay adaptive



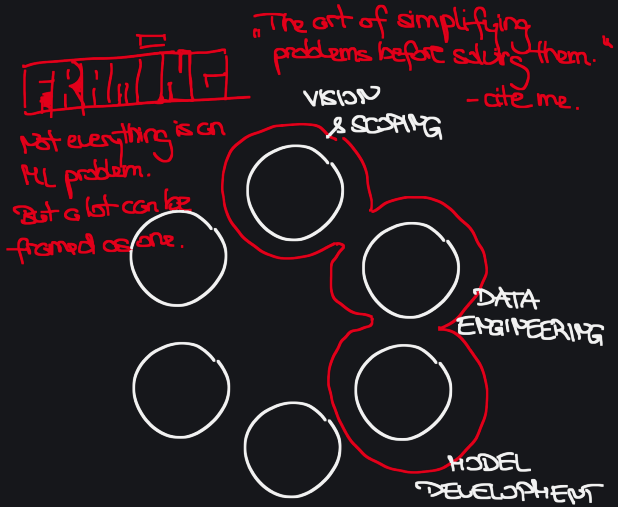
Business analysis closes the loop



4

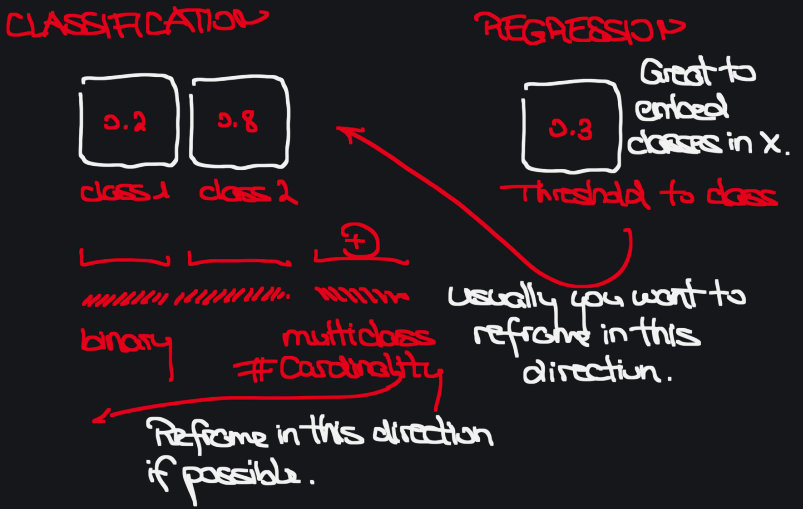
Framing ML Problems

A lot can be framed as an ML problem



Classification versus regression




CLASSIFICATION
0.2
0.8
class 1
class 2
REGRESSION
0.3
Great to embed classes in X.
Threshold to class
binary
multiclass
#Cardinality
Usually you want to reframe in this direction.
Reframe in this direction if possible.

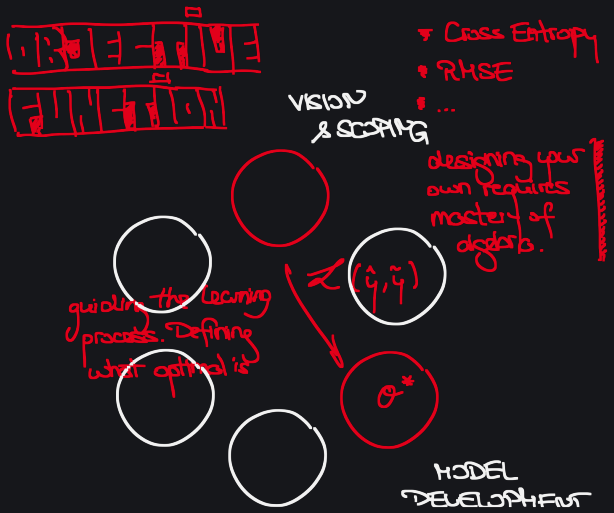
Talk is cheap, sharpen your intuition on MNIST

!! Restrain from „buying" this with an increasing numbes of models.
Hierarchical & Sequential systems are sophisticated but hard to maintain.

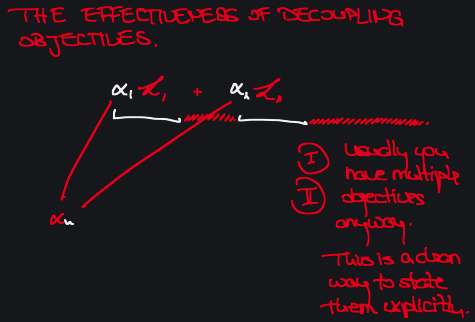


Talk is cheap. Go and try those things on MNIST to sharpen your own intuition.

The objective function defines what optimal is

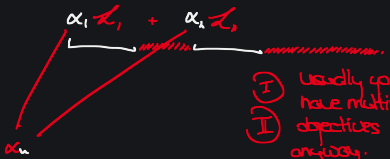


Decoupling objectives



Weight during inference, not during training

THE EFFECTIVENESS OF DECOUPLING OBJECTIVES.



I) Usually you have multiple objectives anyway.

II) This is a clean way to state them explicitly.

Pareto Optimization

Trivial to weight objectives while training.

Everytime you touch is a retrain.

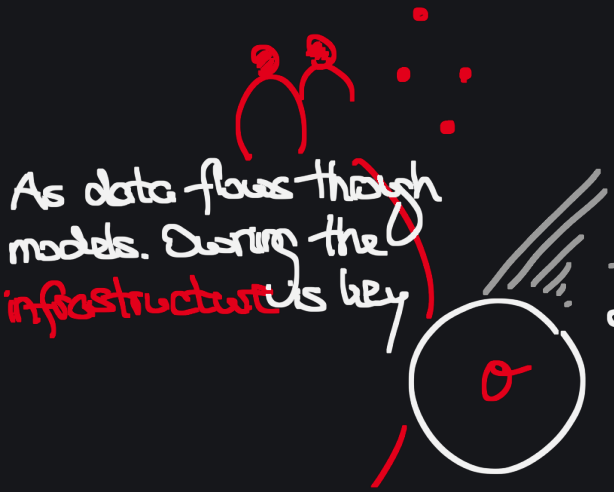
Beautiful opportunity to train two models and adaptively weight during inference.

! Objective Functions require gradients.

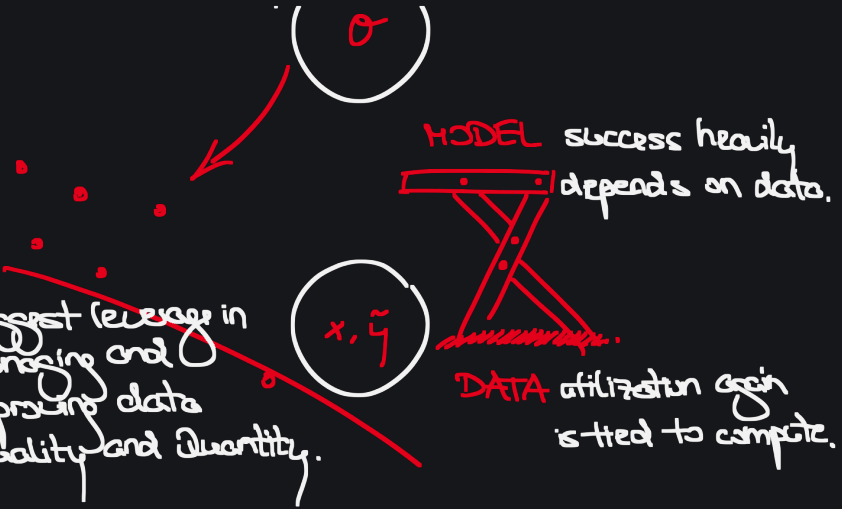
a For everything else we still need metrics.

5

Mind Versus Data


As data flows through models. During the inference it is key
Intelligent architectural designs frozen within the current paradigm.
Data-centric.

Data: model success depends on it



Better data beats better algorithms

"We don't have better algorithms. We just have better data." - Norvig (Director Search @Google)



Remember how more data is a regularization method, driving generalization of your models?

6

Summary

Summary

References

1 Chip Huyen. *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*. Sebastopol, CA: O'Reilly Media, 2022.